Economics 880 – PSet 1

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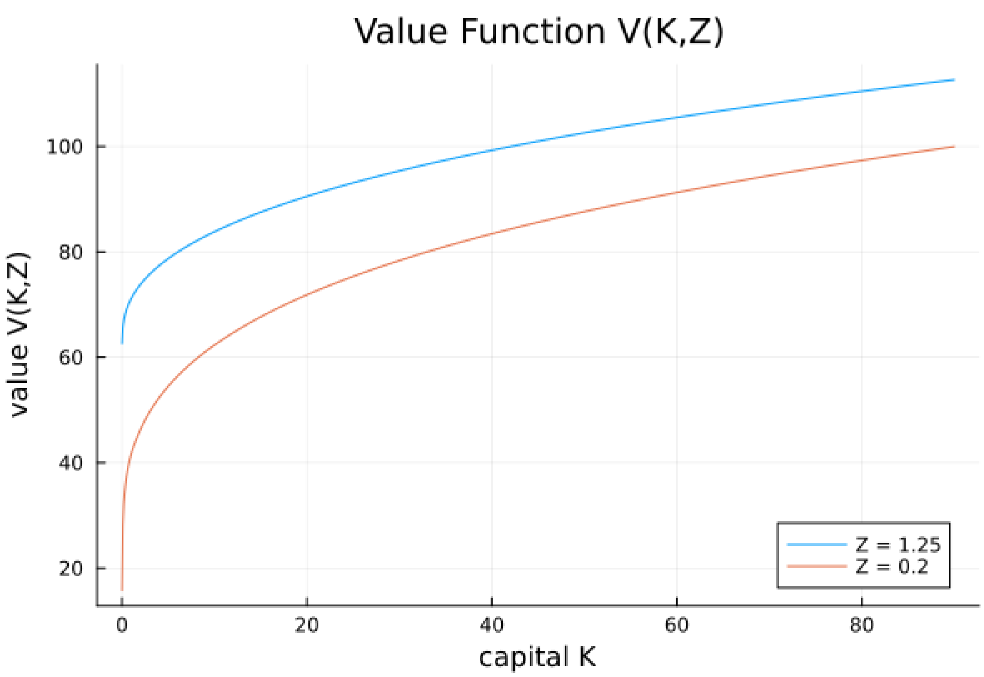
Question 1

The un-parallelized version converged in 1,359 iterations in 1.76 seconds, whereas the parallelized version completed in the same number of iterations but in 13.67 seconds. This result is counter-intuitive, as we would expect that the parallelization would decrease the runtime. However, in this parallelized version, we decided to parallelize over the inner loop that searches over the grid for choice of k' conditonal on (k, z_i). In this way, we lose the ability to use the `choice_lower` variable that exploits the monotonicity of the policy function.

We rewrote the Bellman to instead parallelize over the two technology shock states, while maintaining the use of the `choice_lower` trick, since there will be no race condition parallelizing at the leve of Z. In this version, we have a total runtime of 4.76 which likely reflects the ramp up time of parallelization.

Question 2

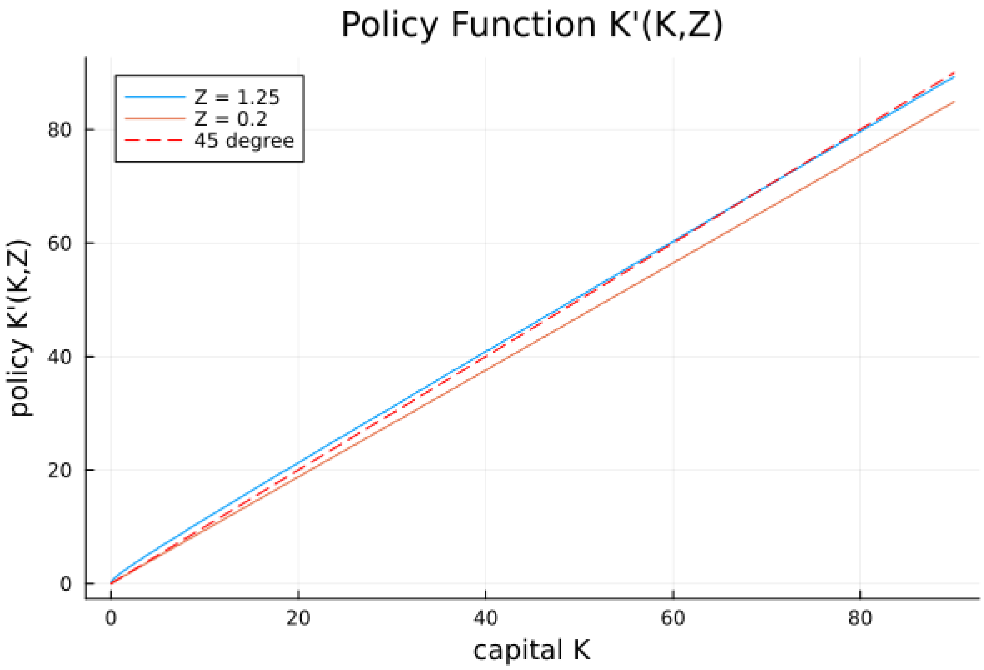
In the figure below, the value function is plotted for conditional on both values of Z. In both cases, the value function is increasing and concave.

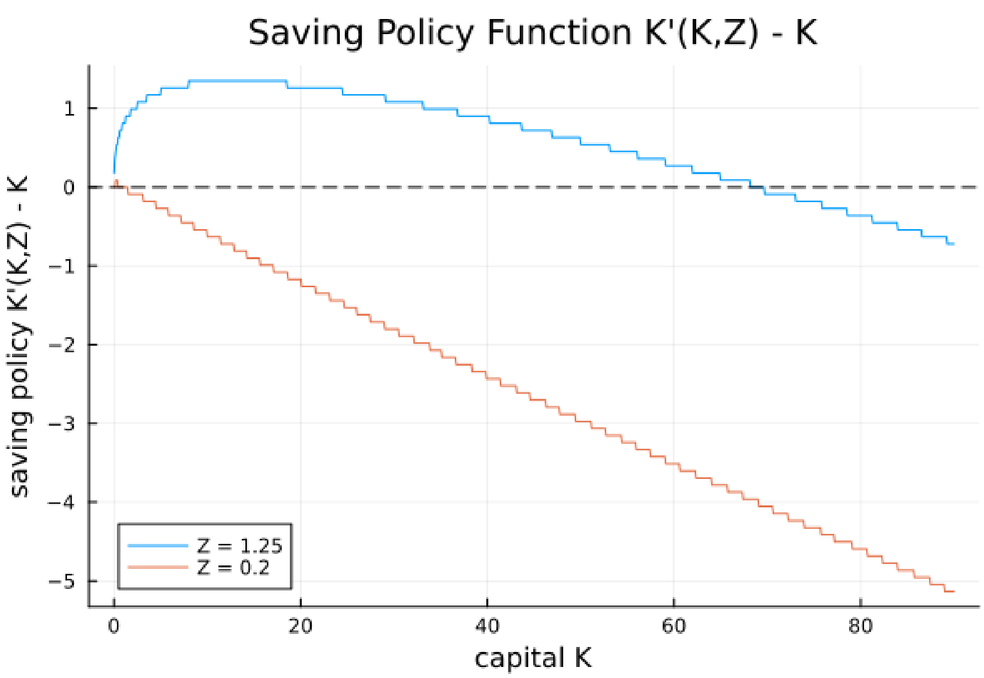


Question 3

In the first figure below, we plot the policy function for both values of the technology shock. For both values, the choice of capital k' is increasing in k. Moreover, the policy function itself is increasing in Z.

In the second figure, we plot saving, defined as $k' - k$, for both values of the technology shock. Saving is generally decreasing in k and increasing in Z. At low levels of capital, the marginal product of capital is high, so the benefit of accumulating additional capital outweighs the cost of foregone consumption. As capital increases, diminishing marginal returns reduce the benefit of additional capital, eventually leading the household to dissave. A higher Z increases the marginal product of capital and therefore increases the incentive to save.





Saving Policy Function K'(K,Z) - K
saving policy K'(K,Z) - K
capital K
Z = 1.25
Z = 0.2
1
0
−1
−2
−3
−4
−5
0
20
40
60
80